Qs:
- Sensory: Repeat w/ light, sharp, and dull?
- After Stereognosis can I verbalize "If abnormal I would perform Graphesthesia and Extinction"?

Neurology Physical Exam

Diagnostic Medicine III

When you conduct the neurologic examination, it is advisable to adopt a fixed routine or examination sequence to minimize omission of one of its important components. Whether you perform a comprehensive or screening examination, organize your thinking into five categories: (1) mental status, speech, and language; (2) cranial nerves (CNs); (3) motor system; (4) sensory system; and (5) reflexes.

STARTING THE EXAM		
Introduces self to patient using first and last name and role. Wash/Sanitize hands Don appropriate PPE Patient is wearing gown, open to back		
General Survey	Verbalize general survey	
Vital Signs	Takes or acknowledges abnormal blood pressure, heart rate, respiratory rate, temperature, weight, height, pain level	
	Set the agenda / Outline the events of the visit	
MENTAL STATUS EVALUATION		
Orientation	Ask patient "What is your name?" 1	
	Ask patient "Where are you right now?" 3	
	Ask pateint "What is today's date?" 2	
	Ask Patient "Why are you here? or What is the reason you are here today?" 4	
MMSE	**Verbalize:** MMSE would be performed as indicated	
CRANIAL NERVES - Verbalize which cranial is being tested		
CN I - Olfactory	Ask patient to occlude one nostril and identify smell bilaterally. Tell them if smell is pleasant/unpleasant	
CN II - Optic	Assess visual acuity using Snellen pocket chart (CN II) at 14 in. Report findings for OU, OD, OS	
	Visual fields by confrontation. *Verbalize* "if abnormal would evaluate each eye separately"	
	Perform a fundoscopic exam bilaterally observing for Optic disc Vessels Retinal Pallor Hemorrhages HOV Road	
CN III, IV, & VI – Oculomotor, Trochlear & Abducens	Inspect size & shape of pupils bilaterally for asymmetry, size	
	Assess direct and consensual pupillary reaction bilaterally using light source	
	Assess extraocular movements bilaterally observing for nystagmus, smooth pursuit, ptosis	H
	Assess pupuillary convergence and lens accomodation bilaterally. Instructing patient to follow your finger to their nose	
CN V – Trigeminal (Sensory & Motor)	Palpate temporal and masseter muscles as patient clenches teeth	
	Ask patient to close eyes. Assess sensation to light touch of the face in V1, V2, V3 distribution	
CN VII – Facial	Observe face for asymmetry	

	Ask patient to: Raise eyebrows, frown, close eyes while you try to open them, Smile showing teeth, puff out cheeks	
CN VIII – Acoustic (bilaterally)	Assess hearing (rubbing thumb and forefinger together 2 inches from each ear or whisper test)	
	Verbalize: if hearing loss is detected, would perform Weber and Rinne test if abnormal hearing was detected	
CN IX & X - Glossopharyngeal & Vagus	Inspect symmetric rise of soft palate and uvula is at midline when patient says "ahh"	
	Verbalize "Test gag reflex using tongue blade" perform on AMS *mainly used in evaluation of patients in a coma	
CN XI – Spinal Accessory	Assess strength sternocleidomastoid strength against resistance bilaterally (head/neck rotation)	
	Assess strength of trapezii againast resistance bilaterally (shoulder shrug)	
CN XII - Hypoglossal	Inspect tongue for fasciculations, or atrophy (*verbalize*)	
	Assess tongue is at midline when patinet sticks out tongue.	
Coordnation / Cerebellar Function		
Coordination & Fine Motor Skills (All tests completed bilaterally)	Assess rapid alternating movements of each hand, finger tap, feet Observing: Speed, Rhythm, Smoothness Dysdiadochokinesia = ipsilateral cerebellar hemisphere dysfx	
	Assess Finger to Nose Test → intention tremor, ipsilateral cerebellar hemisphere lesion (past-pointing) ↳ MS or alchohol abuse	
	Heel to Shin Test → ipsilateral cerebellar dysfunction	
Gait/Station (Stance)	Sit to Stand: Ask pt to rise from chair without use of hands Polymyalgia Rheumatica	
	Have patient walk at least 5 paces to and from you, observing the gait for posture, balance, arm swing, stride PD ASS	
	Tandem gait (heel-toe)	
	Walks on toes	
	Walks on heels	
SENSORY		
Assesses Light Touch, Sharp and Dull, and Temperature Verbalize nerve roots	Ask patient to close eyes and point to where they feel the sensation and if it feels the same bilaterally	
	Assess Light touch	
	Assess Sharp and Dull Touch	
	Would assess temperature using an alcohol pad (*verbalize*)	
	Lateral upper arm (C-5)	
	Lateral forearm, index finger, and thumb (C-6)	
	Middle finger bilaterally (C-7)	
	Medial forearm, ring, and little finger (C-8)	
	Medial upper arm (T-1)	
	Verbalize **:** nipple level is T4	
	Verbalize: umbilical level is T10	
	Proximal 1/3 of thigh (L-1)	
	Middle 1/3 of thigh (L-2)	
	Distal 1/3 of thigh (L-3)	
	Medial lower leg, medial great toe (L-4)	
	Lateral lower leg, toes 2-4 (L-5)	
	Lateral foot, lateral small toe (S-1)	
Assess – Peripheral Neuropathy, MS, Posterior Colum Disease **Proprioception**	Assess proprioception at the DIP joint of the big toe while patient has eyes closed. Perform test bilaterally.	

"Is big toe up or down"

Assess Vibration ↳ B12 deficiency, peripheral neuro, ms	Use the 128 Hz tuning fork to assess vibratory sensation at the 1st MTP joint of the foot while patient has eyes closed. Perform test bilaterally.	
Assess Two-point discrimination - peripheral neuropathy	Have patient close eyes. Use paperclip or calibers to determine two point discrimination on the pad of a finger. "one point or two points" Sensory cortex lesion, Post column Disease	
Stereognosis (bilateral) - contralateral parietal lobe lesion	Have patient close eyes and identify two objects placed in their hand one at a time.	
Graphesthesia (bilateral) - contralateral parietal lobe lesion	Have patient close eyes. Trace a number or letter on the patient's palm. Ask the patient to identify.	
Extinction - contralateral parietal lobe lesion	Have patient close eyes. Touch each arm individually, then simultaneously touch corresponding areas on both arms. Ask where the patient feels your touch with each stimulus. Normally both stimuli are felt. "Both arms, right, or left"	
MUSCLE TONE & STRENGTH		
Inspect *(verbalize)*	Inspecting for muscle for Atrophy, Fasciculations ↳ tremors	
Assess Muscle Tone	Assess muscle tone of upper and lower extremities	
	Verbalize : Increased tone indicates an upper motor neuron lesion. Decreased tone indicates a lower motor neuron lesion.	
Assess Pronator Drift ↳ stroke, ms, brain injury, UMN weakness	Have patient close eyes. Patient holds both arms in front of them at 90 degree angle. Observe for the forearm and palm to drop and turn inward. Then gently tap on the forearm in a downward motion observing for the arm to return to baseline. UMN lesion, stroke, tumors	
Assess Muscle Strength bilaterally *Verbalize the Nerve root*	Shoulder shrug	
	Shoulder abduction (C5, C6)	
	Elbow flexion (C5, C6) and extension (C6, C7, C8)	
1.	Wrist extension (C6,C7,C8, Radial Nerve)	
3.	Digit abduction (C8,T1, Ulnar Nerve)	
4.	Thumb abduction (C8, T1, Median Nerve)	
2.	Digit extension(C7, C8, radial nerve)	
	Hip flexion (L2, L3, L4) and Hip Extension (S1)	
	Hip adduction (L2, L3, L4)	
	Hip Abdunction (L4, L5, S1)	
	Knee flexion (L5, S1, S2) and extension (L2, L3, L4)	
	Ankle dorsiflexion (L4, L5) and Plantar Flexion (S1)	
DEEP TENDON REFLEXES		
Assess DTR Bilaterally	Biceps Reflex (C5,C6)	
	Triceps Reflex (C6,C7)	
	Brachioradialis Reflex (C5,C6)	
	Patellar Reflex (L2, L3, L4)	
	Ankle Jerk (achillies tendon) (S1)	
SPECIAL TESTS		
Ankle Clonus ↳ UMN lesion, stroke, ms, cerebral palsy, TBI	With your other hand, dorsiflex and plantar flex the foot a few times while encouraging the patient to relax, then sharply dorsiflex the foot and maintain it in dorsiflexion. Obesrve for rhythmic oscillations	
Brudzinski Sign (meningitis)	Patient is supine. Passively flex chin toward sternum observing for flexion of hips and knees	
Kernig Sign (meningitis)	Position the patient supine with hip and knee flexed at 90 degrees. Passively extend the knee. kick knee up so sole faces sky	

Asterixis → uremia **(encephalopathy)**	Ask the patient to "stop traffic" by extending both arms, with wrists dorsiflexed and fingers spread. Observe for abnormal "flapping" tremor at the wrist	
Straight Leg Raise ↳ Sciatic pain, Herniation, Stenosis, Radiculopathy	Place the patient in the supine position. Raise the patient's relaxed and straightened leg, flexing the thigh at the hip Document at what angle pain and side low back pain occurs	
Plantar Reflex (Babinski) → UMN · stroke, Tumor, MS	Stroke the lateral aspect of the sole from the heel to the ball of the foot, curving medially across the ball observing for <u>dosiflexion</u> of the big toe plantarflexion in babies	
Romberg test ↳ sensory ataxia, peripheral neuropathy, MS, vestibular dysfx	Patient stands with both feet together, arms at side, with eyes open. If stable, ask patient to close their eyes and continue to stand for 20-30 sec. Observe for loss of balance. Provider guards patient.	

Order:
- Brudzinski
- Kernig
- Straight leg Raise
- Asterixis
- Ankle clonus
- Babinki's sign
- Romberg

Supine

Sitting